

Culinary Corridors: Mapping Where Food Similarity Breaks Distance

Committee-Ready Final Report for Fisher Award Review

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Abstract

Culinary Corridors uses geographic information science to ask when cuisine similarity follows distance and when it breaks distance-based expectation. Cuisines are represented as filtered ingredient profiles, compared through pairwise similarity metrics, anchored to approximate geography, and modeled against distance. The central object is the residual: observed cuisine similarity minus distance-predicted similarity. Positive residuals become candidate culinary corridors. The global model is a discovery screen, East/Southeast Asia is the primary focused case, and residual bridge scores translate pairwise links into mapped place-level roles. Run 4 v2 adds a corridor-accessibility proxy to test whether the strongest East/Southeast Asian residuals are spatially interpretable through subregion, distance, coastal/island position, maritime context, and transparent barrier flags. The project does not claim global representativeness or causal proof. Its contribution is methodological and cartographic: GIS transforms recipe similarity into spatial evidence about corridors, bridges, boundaries, and residual geography.

1 Executive Summary

The project asks: **when does cuisine similarity follow geographic distance, and when does it break distance-based expectation?** The prototype shows that distance matters, but does not explain all cuisine similarity. The strongest residuals become candidate culinary corridors. The most defensible interpretation comes from a focused East/Southeast Asia case, and the clearest Fisher-style geospatial insight comes from residual bridge scores and the Run 4 v2 corridor-accessibility proxy.

The final scope is deliberately disciplined: a global discovery screen, an East/Southeast Asia primary case, and an Iberian/Atlantic-Pacific secondary or diagnostic case. Mechanism language remains cautious. Positive residuals are consistent with possible histories of exchange, but they are not proof of migration, trade, colonialism, maritime movement, or terrain causality.

2 Research Question and Problem

Food is a scientific and cultural system with global reach, but food similarity is not evenly distributed across space. Similar ingredient repertoires may emerge through proximity, agriculture, ecology, exchange, diaspora, shared food technologies, colonial history, platform bias, or pantry overlap. A non-spatial analysis can cluster cuisines by ingredients, but it cannot say whether that similarity is spatially expected or spatially surprising.

The research question is therefore:

When does cuisine similarity follow geographic distance, and when does it break distance-based expectation in ways that reveal candidate culinary corridors, bridge regions, or boundary patterns?

3 Why the Project Fits the Fisher Award

The project is Fisher-aligned because GIS is the engine of the finding rather than a display layer. The key evidence requires coordinates, distance, residual geography, focused spatial cases, and mapped bridge roles. Without GIS, the project is ingredient clustering. With GIS, it becomes an analysis of where cuisine follows distance, where it breaks distance, and which places become residual bridge nodes.

The internal Run 4 scorecard, based on the uploaded Fisher strategy packet, improved the package from 80.8 to 91.9 out of 100. This is not an official Fisher score; it is a project-specific diagnostic showing that the revised package better satisfies innovation, GIS use, data documentation, analytical execution, and visualization criteria.

4 Data

The data stack combines a cuisine-labeled recipe/ingredient corpus, ingredient normalization, a generic-ingredient filtering policy, approximate cuisine-to-geography anchors, pairwise distances, cosine and Jaccard similarity metrics, residual corridor outputs, focused case outputs, residual bridge scores, and boundary/permeability summaries.

The limitations are visible by design. The recipe corpus is a proxy rather than a census of world cuisine. Cuisine labels are not exact countries or homogeneous cultures. Ingredient normalization is interpretive. The geographic anchors are approximate. Positive residuals are model outputs, not historical proof.

5 Methods

5.1 Ingredient profiles

Recipes are transformed into cuisine-by-ingredient vectors. Generic/pantry terms are removed or down-weighted through a documented filtering policy so that similarity is less dominated by platform vocabulary.

5.2 Similarity metrics

Cuisine pairs are compared with cosine similarity and robustness metrics such as Jaccard similarity. This creates a non-spatial ingredient-similarity baseline.

5.3 Geographic distance model

Cuisines are mapped to approximate coordinates. For each cuisine pair, the model compares ingredient similarity to log geographic distance:

$$\text{similarity}_{ij} = \alpha + \beta \log(\text{distance}_{ij}) + \epsilon_{ij}.$$

The filtered Run 2 v2 model used 190 cuisine dyads. The log-distance coefficient was -0.1158 and the model R-squared was 0.3553. Distance explains a meaningful share of similarity, but leaves residual structure for GIS analysis.

5.4 Residual culinary corridors

The central quantity is:

$$\text{residual}_{ij} = \text{observed similarity}_{ij} - \text{distance-predicted similarity}_{ij}.$$

Positive residuals are candidate culinary corridors: pairs more similar than distance alone predicts.

5.5 Focused cases and geospatial-only analyses

The global analysis is a discovery layer. East/Southeast Asia carries the primary focused inference. Additional geospatial-only analyses include residual bridge scores, boundary/permeability groupings, and the Run 4 v2 corridor-accessibility proxy.

6 Core Findings

6.1 Distance matters, but residuals remain meaningful

Cuisine similarity decreases with log distance, but not completely. This supports the main spatial question: where does food similarity break distance?

6.2 East/Southeast Asia is the strongest focused case

The top residuals in the East/Southeast Asia focused case include:

Cuisine A	Cuisine B	Residual	Distance km
thai	vietnamese	0.359	808
chinese	korean	0.306	2118
filipino	thai	0.219	2262
filipino	vietnamese	0.209	1465
chinese	filipino	0.133	3103

These links are spatial associations, not causal explanations.

6.3 Run 4 v2 adds corridor-accessibility context

The Run 4 v2 enhancement creates a transparent corridor/accessibility proxy using same-subregion status, distance, coastal/island context, maritime context, and conservative barrier flags.

Cuisine A	Cuisine B	Accessibility	Residual-alignment
thai	vietnamese	0.880	0.316
chinese	korean	0.757	0.231
filipino	vietnamese	0.862	0.180
filipino	thai	0.800	0.176
chinese	filipino	0.543	0.072

The strongest residuals, especially Thai-Vietnamese and Chinese-Korean, also sit in high-accessibility corridor contexts. Filipino links illustrate why maritime and archipelagic positioning matters for the focused case. This is a proxy, not a measured terrain or route model.

6.4 Bridge scores are the clearest spatial-necessity result

The bridge index aggregates residual links into mapped place-level roles:

Cuisine	Bridge index	Spatial role
filipino	0.863	primary_case_bridge
russian	0.839	global_residual_bridge
southern_us	0.692	global_residual_bridge
jamaican	0.686	global_residual_bridge
french	0.653	context_bridge_role
spanish	0.532	secondary_corridor_or_diagnostic

The bridge index cannot be obtained from ingredient vectors alone because it depends on the distance baseline and spatial residual network.

7 Figure Walkthrough

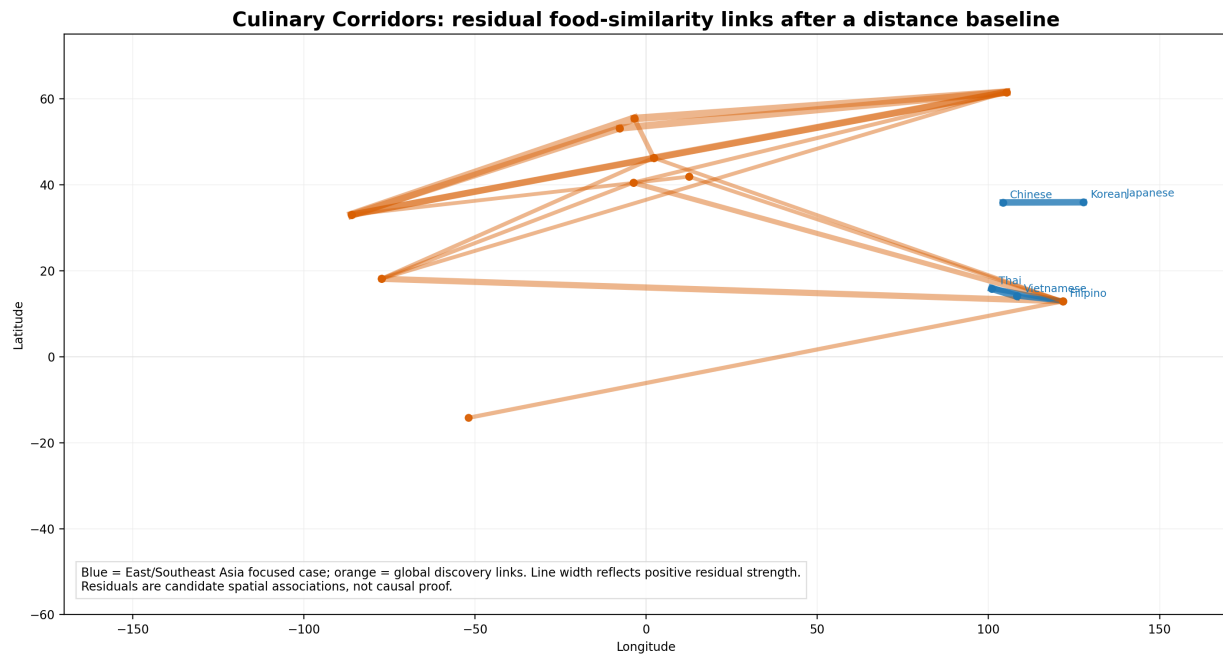


Figure 1: Hero spatial argument. Filtered positive residual culinary corridors after a geographic distance baseline. This map shows where cuisine similarity exceeds distance-based expectation. It is a discovery figure, not causal proof.

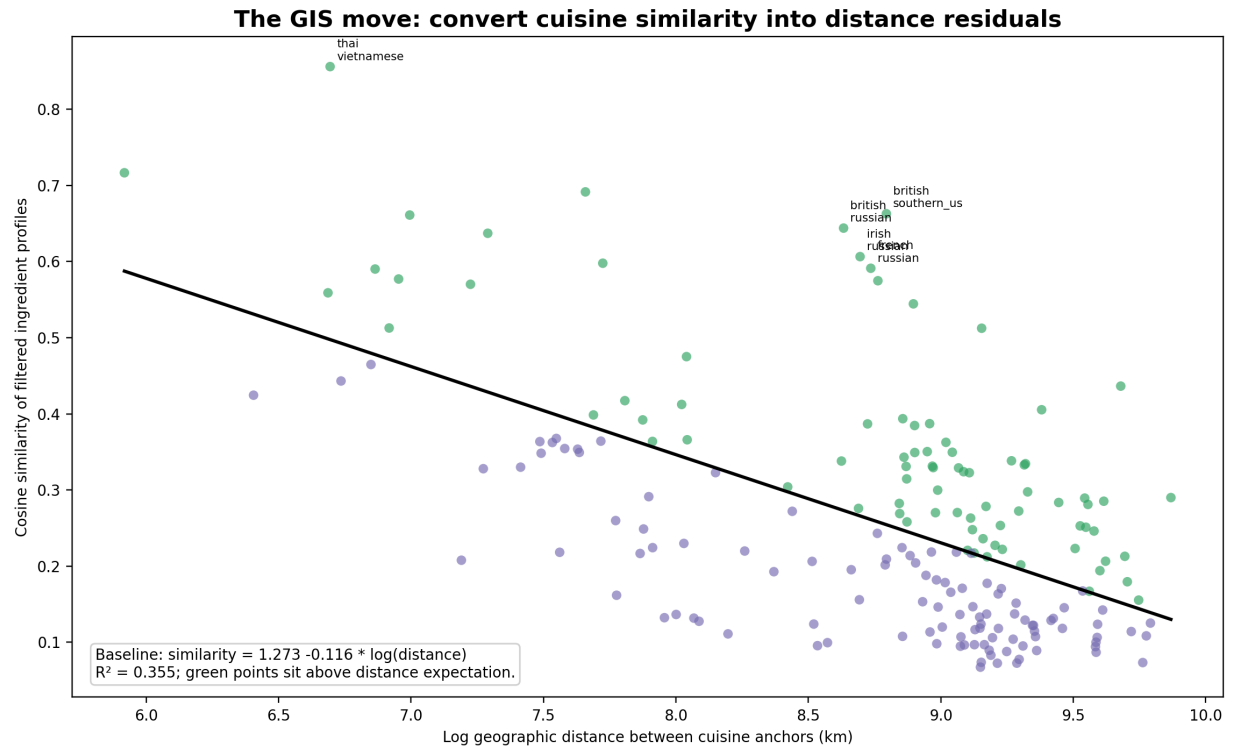


Figure 2: Residual method figure. The distance trend defines spatial expectation; points above the trend become positive residual culinary corridors.

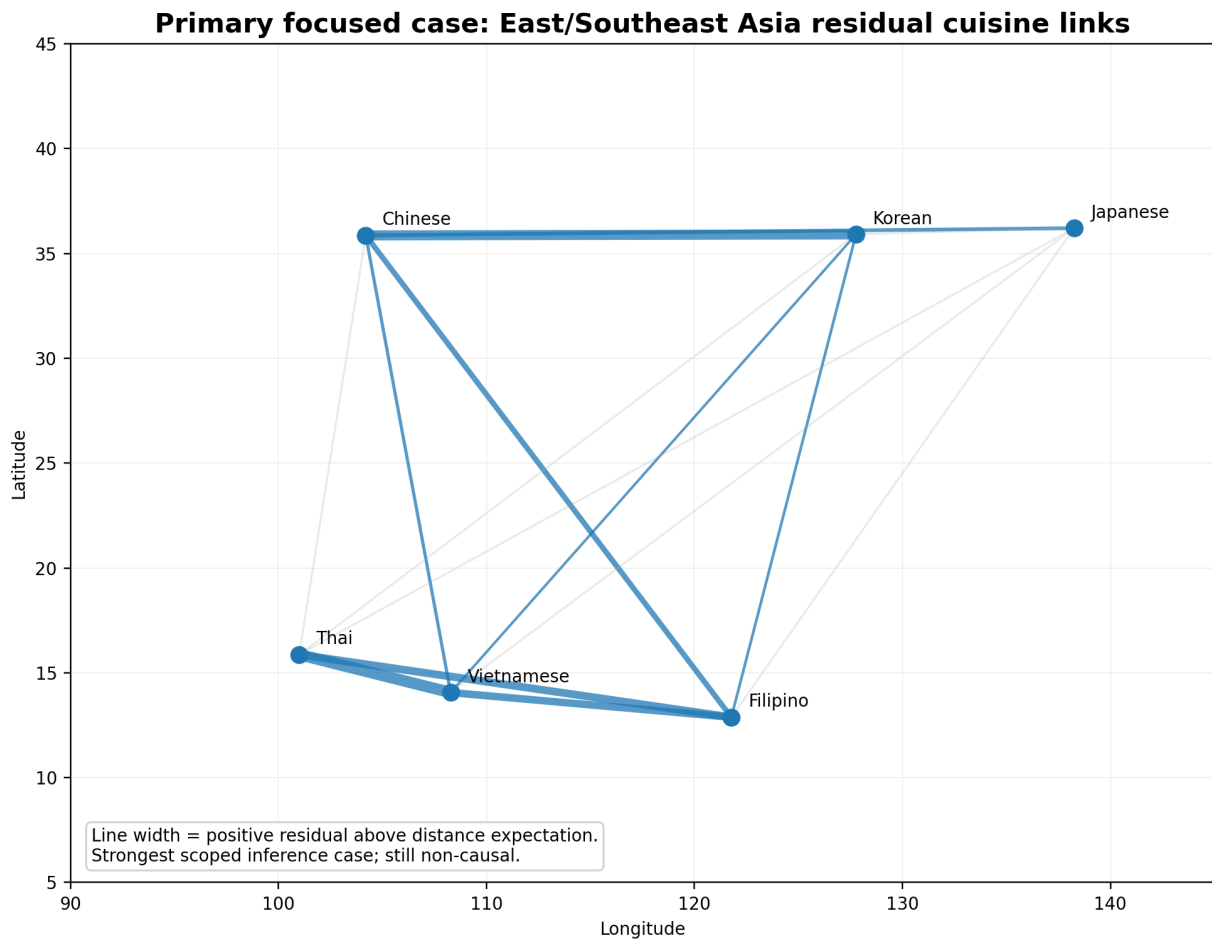


Figure 3: East/Southeast Asia focused case. This is the strongest inference case because it is spatially coherent and less dependent on broad global interpretation.

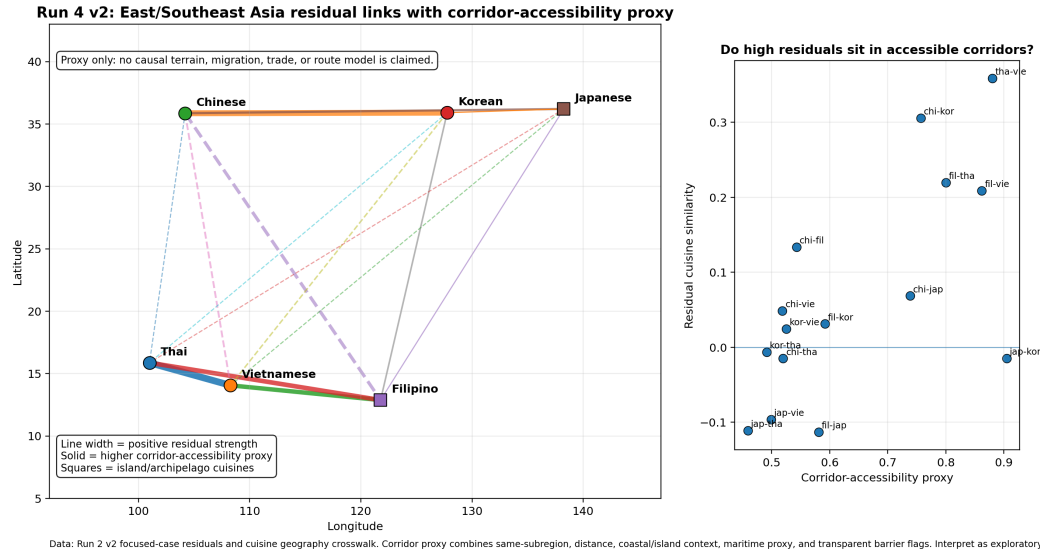


Figure 4: Run 4 v2 corridor-accessibility enhancement. The figure compares East/Southeast Asia residual links with a corridor/accessibility proxy based on subregion, distance, coastal/island context, maritime context, and barrier flags. It is not a measured topographic cost surface.

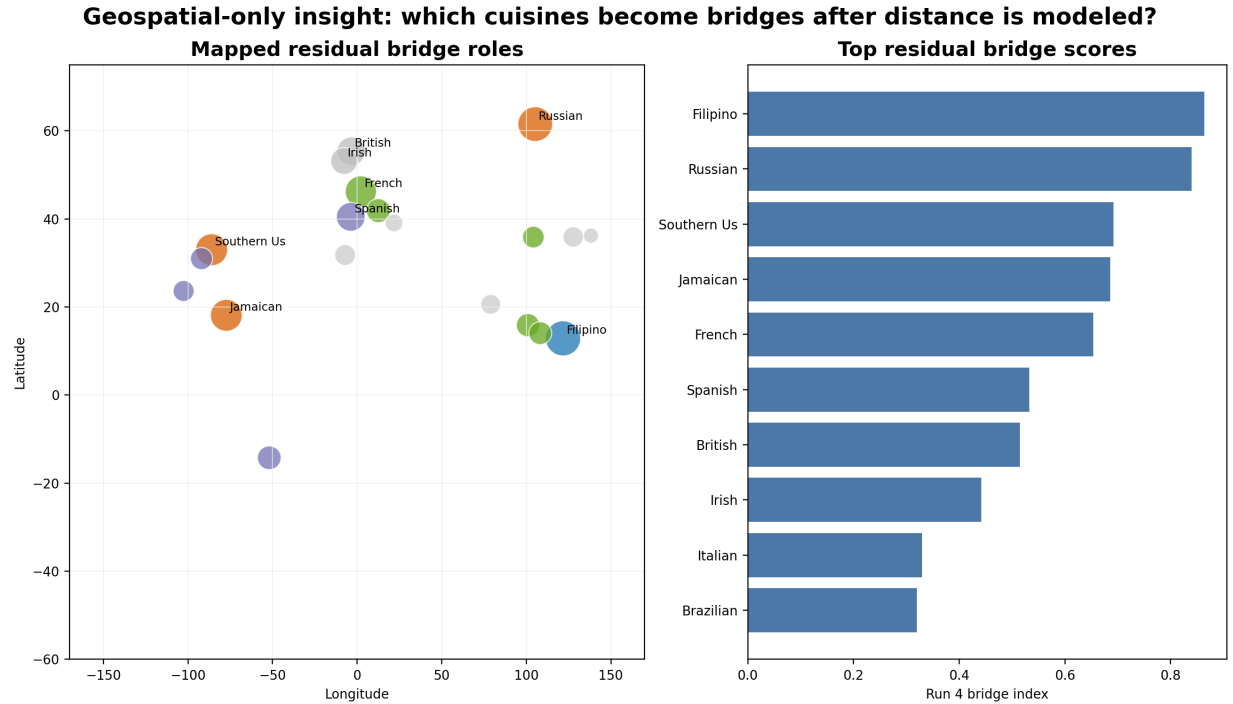


Figure 5: Residual bridge-index figure. This visual converts pairwise residual similarity into mapped place-level roles, demonstrating a result that requires GIS.

8 Limitations

- The recipe corpus is a platform-mediated proxy, not a globally representative sample.

- Cuisine labels are approximate and not equivalent to nation-states.
- Ingredient normalization and generic-ingredient filtering affect the result.
- Positive residuals are associations, not causal evidence.
- Run 4 v2 uses a corridor/accessibility proxy, not measured topography or least-cost routing.
- Global findings should be treated as discovery; focused cases carry the stronger interpretation.

9 Submission Recommendation

The recommended Fisher submission format is a map-led StoryMap with a static report/PDF backup. The recommended figure sequence is hero map, residual method, East/Southeast Asia focused case, Run 4 v2 corridor-accessibility map, residual bridge index, and limitations/source panel.

10 Conclusion

Culinary Corridors turns food similarity into spatial evidence. It asks where cuisine follows distance, where it breaks distance, and which places become bridge nodes in residual geography. The project is strongest when it stays disciplined: global discovery, focused East/Southeast Asia inference, explicit data caveats, and cautious mechanism language.

11 Selected Source and Artifact Notes

Internal source artifacts include the Run 2 v2 filtered distance baseline summary, the Run 2 v2 geospatial analysis summary, the Run 4 revised thesis memo, the Run 4 geospatial upgrade summary, the Run 4 data/source risk review, and the Run 4 claim/compliance audit. Public source families used in the broader project package include Harvard Center for Geographic Analysis Fisher Prize pages, Natural Earth base map documentation, UN M49 regional classifications, UN/FAO-style data portals in the source register, and the recipe-corpus source notes documented in the project manifests.